



Industrial Internship Report on UrbanFlowAI

Smart City Traffic Forecasting using Machine Learning

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "UrbanFlow AI – Smart City Traffic Forecasting using Machine Learning," an intelligent web-based application developed to predict traffic congestion levels using historical traffic and environmental data. The project leverages Machine Learning algorithms to analyze factors such as temperature, weather conditions, holidays, day of the week, hour of the day, and month to forecast traffic volume accurately. A user-friendly dashboard was built using Streamlit, allowing users to select different cities and visualize traffic predictions along with interactive charts and analytics. The application also features dynamic city background images, real-time prediction capability, and an enterprise-style interface, making it both practical and visually engaging. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.



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1.Preface

The six-week Industrial Internship organized by **upskill Campus (USC)** in collaboration with **UniConverge Technologies (UCT)** and **The IoT Academy** provided me with an excellent opportunity to gain practical exposure to modern technologies and industry-oriented project development. This internship helped me bridge the gap between theoretical concepts learned in academics and their real-world implementation through hands-on experience in Machine Learning, Python programming, data analysis, and web application development.

As part of this internship, I developed "**UrbanFlow AI – Smart City Traffic Forecasting using Machine Learning**," an intelligent web-based application designed to predict traffic congestion using Machine Learning techniques. The system analyzes various parameters such as temperature, weather conditions, holidays, day of the week, hour, and month to estimate traffic levels. A modern dashboard was developed using Streamlit, featuring interactive visualizations, city-specific backgrounds, analytical charts, and real-time traffic predictions. The project demonstrates how Artificial Intelligence can contribute to efficient traffic management and support smart city initiatives.

The internship program was systematically planned with structured learning modules, practical assignments, project implementation, documentation, and GitHub-based project submission. Throughout these six weeks, I enhanced my knowledge of Python, Machine Learning, data preprocessing, visualization techniques, Streamlit application development, Git and GitHub, and technical documentation. In addition to technical skills, I also improved my logical thinking, debugging ability, project management, and self-learning capabilities.

I would like to express my sincere gratitude to **upskill Campus**, **UniConverge Technologies Pvt. Ltd.**, and **The IoT Academy** for providing this valuable learning opportunity. I am thankful to all the mentors, trainers, coordinators, faculty members, and my college for their continuous guidance and support throughout the internship.

Finally, I would like to encourage my juniors and peers to actively participate in internship programs and continuously explore new technologies.



2.Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





FACTORY WATCH

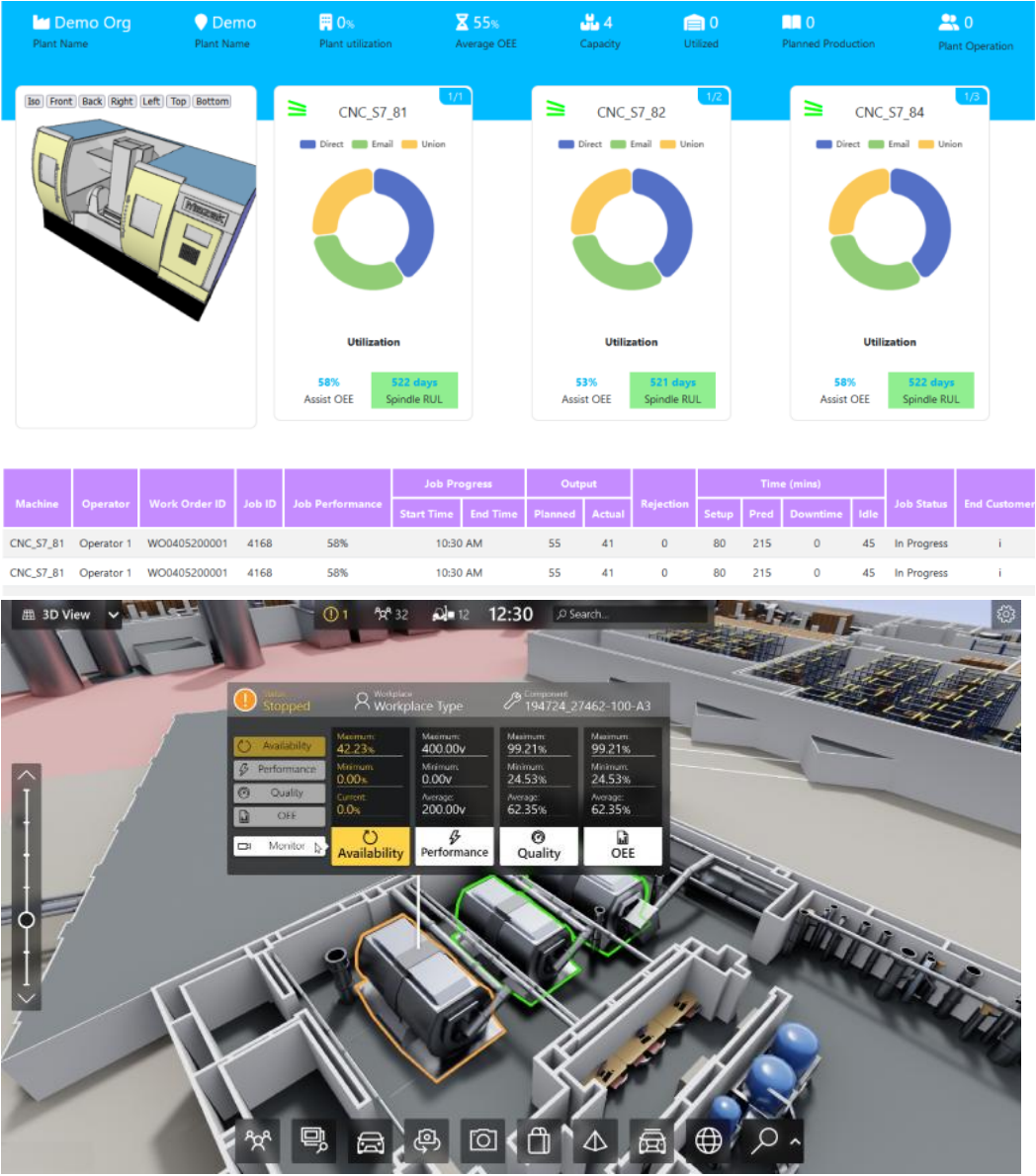
ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



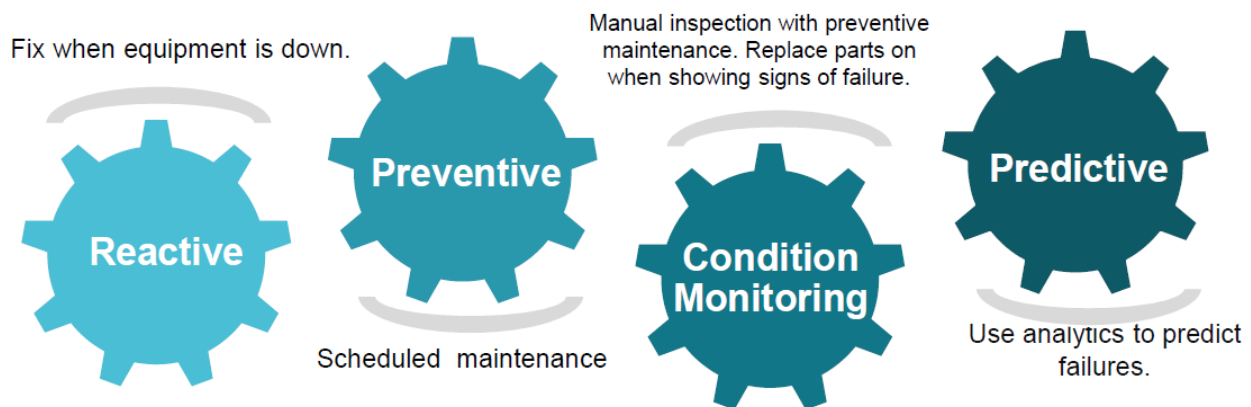


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

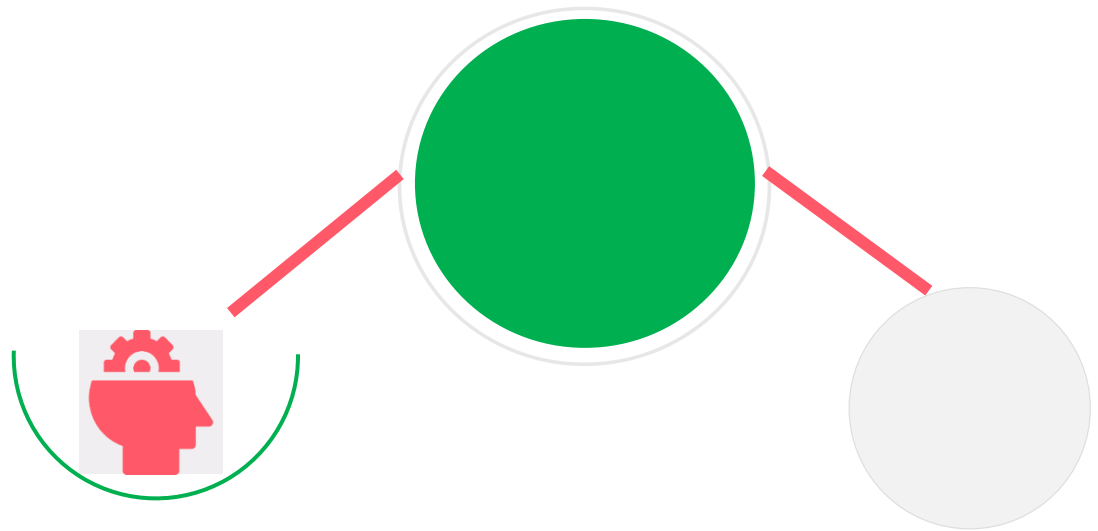
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

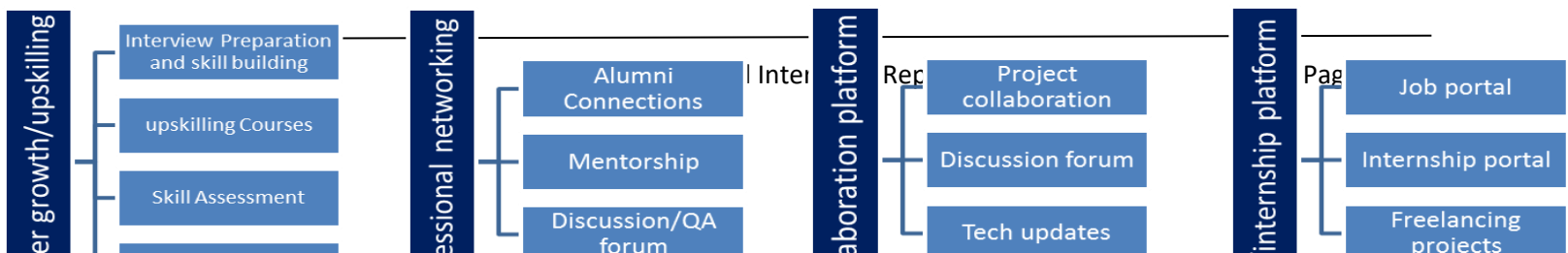
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>





2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ▣ get practical experience of working in the industry.
- ▣ to solve real world problems.
- ▣ to have improved job prospects.
- ▣ to have Improved understanding of our field and its applications.
- ▣ to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Python Software Foundation, **Python Documentation**, <https://docs.python.org/>
- [2] Streamlit Inc., **Streamlit Documentation**, <https://docs.streamlit.io/>
- [3] Scikit-learn Developers, **Scikit-learn Documentation**, <https://scikit-learn.org/>



2.6 Glossary

Terms	Description
Random Forest	A Machine Learning algorithm that combines multiple decision trees to improve prediction accuracy.
Machine Learning	A branch of AI where computers learn patterns from data to make predictions.
API	A set of rules that allows different software applications to communicate with each other.
Feature Engineering	The process of selecting and transforming input variables to improve Machine Learning model performance.
Streamlit	A Python framework used to build interactive web applications.
Model Training	The process of teaching a Machine Learning model using historical data to recognize patterns and make predictions.



3.Problem Statement

SmartCityTrafficForecasting

With the rapid growth of urban populations and the increasing number of vehicles on roads, traffic congestion has become one of the major challenges faced by modern cities. Traffic jams lead to increased travel time, higher fuel consumption, air pollution, road accidents, and economic losses. Traditional traffic monitoring systems mainly provide current traffic information but often lack the ability to predict future traffic conditions accurately, making proactive traffic management difficult.

The objective of the **SmartCityTrafficForecasting** project is to develop an intelligent Machine Learning-based system capable of predicting traffic volume using historical traffic and environmental data. The system analyzes important factors such as temperature, weather conditions, holidays, day of the week, hour of the day, and month to estimate future traffic levels. By identifying traffic patterns in advance, the system helps authorities and commuters make informed decisions regarding traffic management, route planning, and resource allocation.

To address this problem, a Machine Learning model based on the **Random Forest Regressor** algorithm was developed and integrated with an interactive **Streamlit** dashboard. The application provides real-time traffic predictions, visual analytics, and dynamic city-specific interfaces, enabling users to understand traffic trends easily. The proposed solution demonstrates how Artificial Intelligence and Machine Learning can be applied to improve urban transportation systems and contribute to the development of smarter, safer, and more efficient cities.



4.Existing and Proposed solution

Existing Solution

Traffic management systems currently used in many cities primarily rely on GPS navigation, traffic cameras, and sensor-based monitoring to provide real-time traffic information. Popular navigation applications such as Google Maps and Waze use live user data and GPS signals to estimate current traffic conditions and suggest alternative routes. Intelligent Transportation Systems (ITS) also use cameras, IoT sensors, and traffic control centers to monitor vehicle movement and manage traffic flow.

Although these systems are effective in displaying current traffic conditions, they have several limitations:

- They mainly focus on **real-time monitoring** rather than predicting future traffic conditions.
- Most systems require continuous internet connectivity and access to live GPS or sensor data.
- They often involve expensive infrastructure such as cameras, sensors, and IoT devices.
- They provide limited insights into how environmental factors such as weather, holidays, or seasonal variations influence traffic patterns.
- Many existing solutions do not offer customizable dashboards for detailed traffic analytics and visualization.

Due to these limitations, there is a need for an intelligent system capable of forecasting traffic conditions using historical data and Machine Learning techniques.

Proposed Solution

- To overcome the limitations of existing systems, the **SmartCityTrafficForecasting** project proposes an AI-powered traffic prediction system using Machine Learning. The application is designed to predict traffic volume based on historical traffic data and multiple influencing factors such as temperature, weather conditions, holidays, day of the week, hour of the day, and month.
- The project uses the **Random Forest Regressor** algorithm to train the prediction model and provides an interactive web dashboard developed using **Streamlit**. Users can enter traffic-related parameters, select a city, and instantly receive predicted traffic levels along with visual analytics. The dashboard also includes interactive graphs, city-specific background images, and an intuitive user interface, making it suitable for traffic analysis and demonstration purposes.

Value Addition

The proposed system offers several improvements over conventional traffic monitoring solutions:

- Predicts future traffic conditions instead of only displaying current traffic status.
- Utilizes Machine Learning to improve prediction accuracy using historical traffic patterns.
- Provides an interactive and user-friendly dashboard developed with Streamlit.
- Includes graphical analysis through interactive Plotly visualizations for better understanding of traffic trends.
- Supports multiple cities with dynamic city-specific backgrounds to enhance the user experience.
- Assists city administrators, urban planners, and commuters in making informed traffic management decisions.
- Offers a lightweight and cost-effective solution that can be expanded by integrating real-time traffic APIs, weather APIs, IoT sensors, and cloud deployment in the future.



4.1 Code submission (Github link)-

<https://github.com/Maskeenluthra12/upskillcampus/tree/main/SmartCityTrafficForecasting>

4.2 Report submission (Github link) :

https://github.com/Maskeenluthra12/upskillcampus/blob/main/reports/SmartCityTrafficForecasting_Maskeen_USC_UCT.pdf



5 Proposed Design/ Model

The **SmartCityTrafficForecasting** system is designed to predict traffic volume using Machine Learning by analyzing historical traffic data and environmental factors. The proposed solution follows a systematic workflow consisting of data collection, data preprocessing, model development, and prediction through an interactive web dashboard.

Initially, a structured traffic dataset containing features such as **temperature, weather conditions, holiday status, hour of the day, day of the week, month, and traffic volume** was collected. The dataset was then preprocessed by handling missing values, encoding categorical variables using Label Encoding, selecting relevant features, and splitting the data into training and testing sets to prepare it for Machine Learning.

A **Random Forest Regressor** algorithm was selected for model development due to its high prediction accuracy, ability to handle complex relationships, and robustness against overfitting. The model was trained using historical traffic data and evaluated using standard performance metrics such as **Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score**. After successful evaluation, the trained model and label encoders were saved using the Joblib library for future predictions.

The trained model was integrated into a **Streamlit-based web application**, where users can provide input parameters including city, temperature, weather condition, holiday status, hour, day, and month. Based on these inputs, the system predicts the expected traffic volume in real time and displays the results through an interactive dashboard. The dashboard also includes traffic analytics, hourly traffic trends, weather-wise analysis, holiday comparisons, monthly trends, and dynamic city-specific background images, providing a visually appealing and user-friendly experience.

Overall, the proposed design combines Machine Learning with an interactive web interface to provide an efficient, accurate, and scalable traffic forecasting solution. The system can further be enhanced by integrating live traffic APIs, IoT sensor data, weather forecasting services, and cloud deployment to support real-time smart city traffic management.



5.1 High Level Diagram

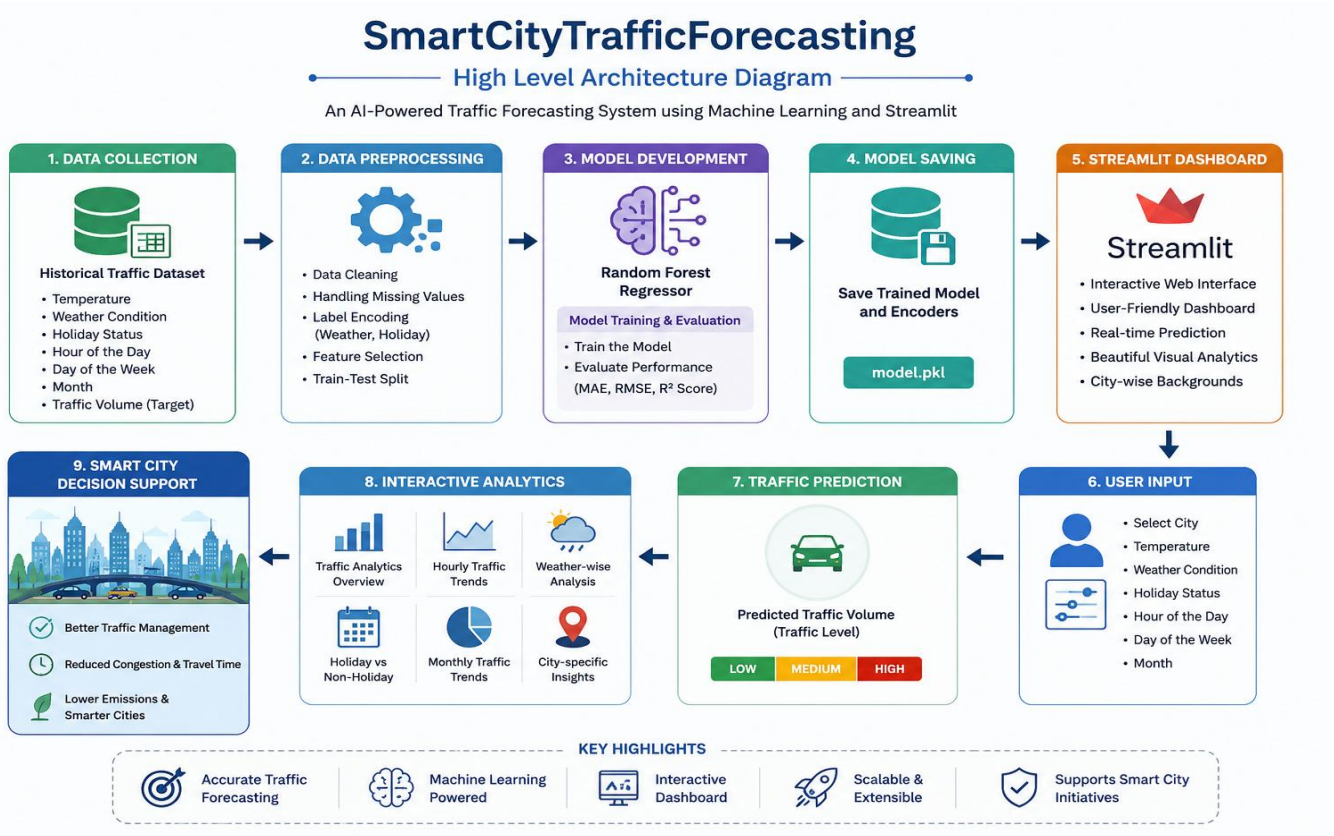
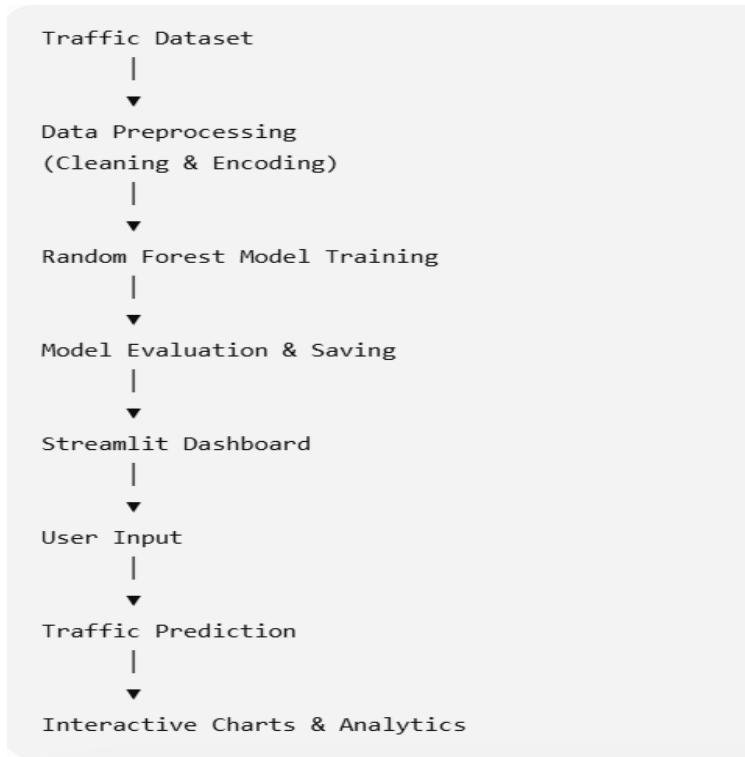
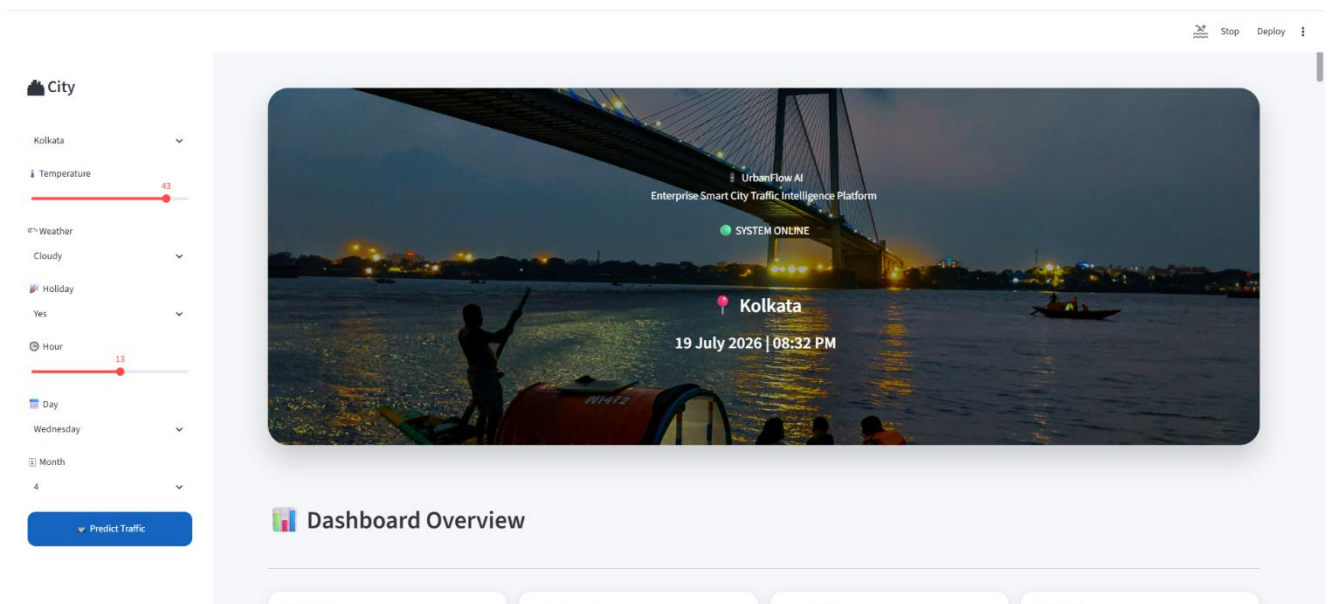


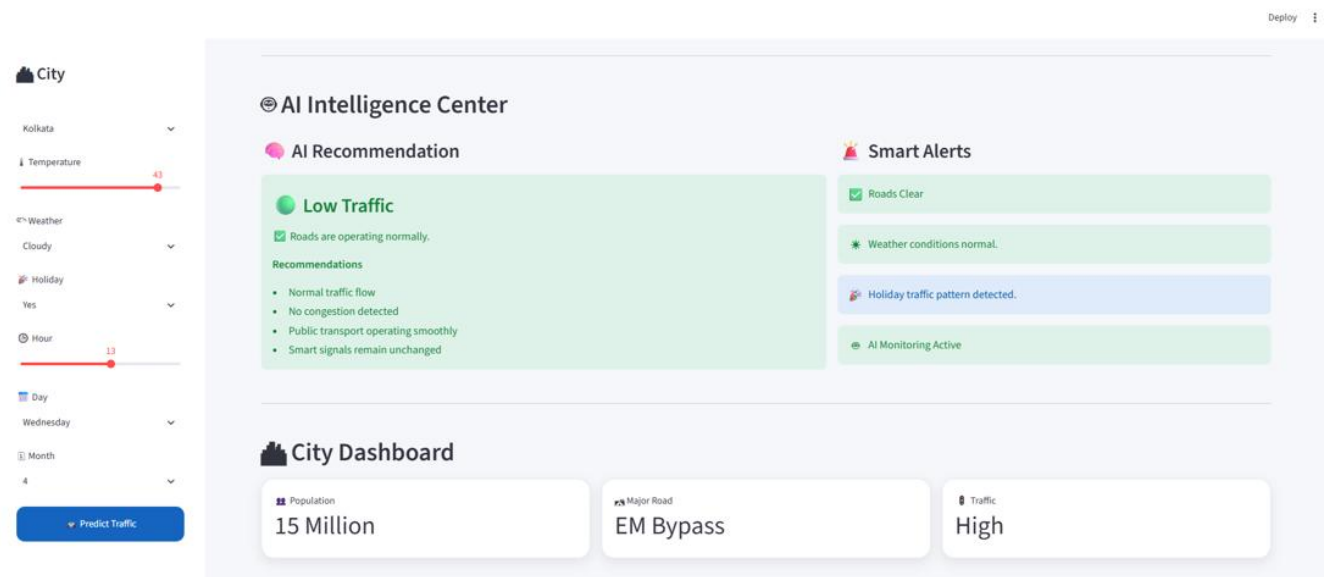
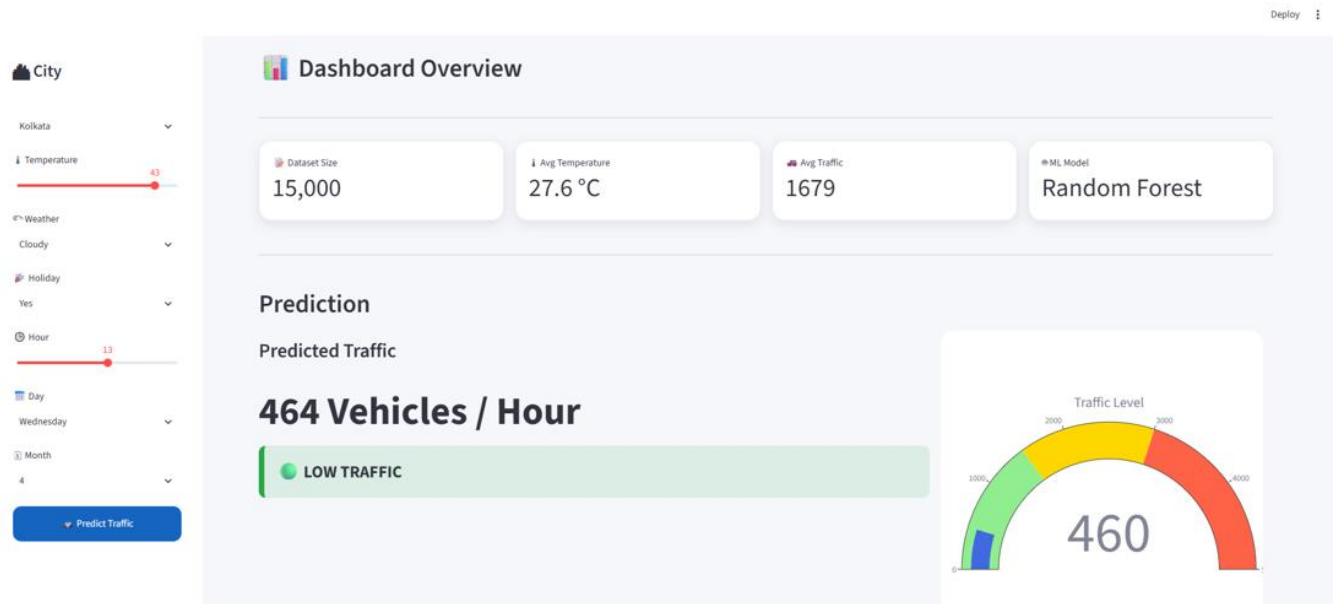
Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram



5.3 Interfaces





City

Kolkata

Temperature 43

Weather Cloudy

Holiday Yes

Hour 13

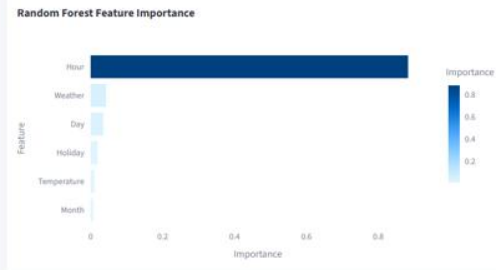
Day Wednesday

Month 4

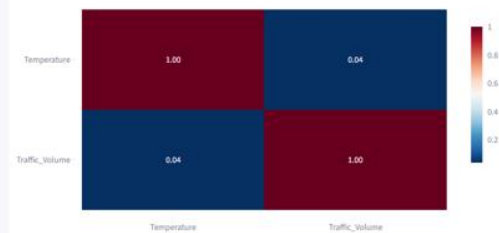
Predict Traffic

Model Analytics

Feature Importance



Correlation Matrix



Dataset Explorer

Number of rows

City

Kolkata

Temperature 43

Weather Cloudy

Holiday Yes

Hour 13

Day Wednesday

Month 4

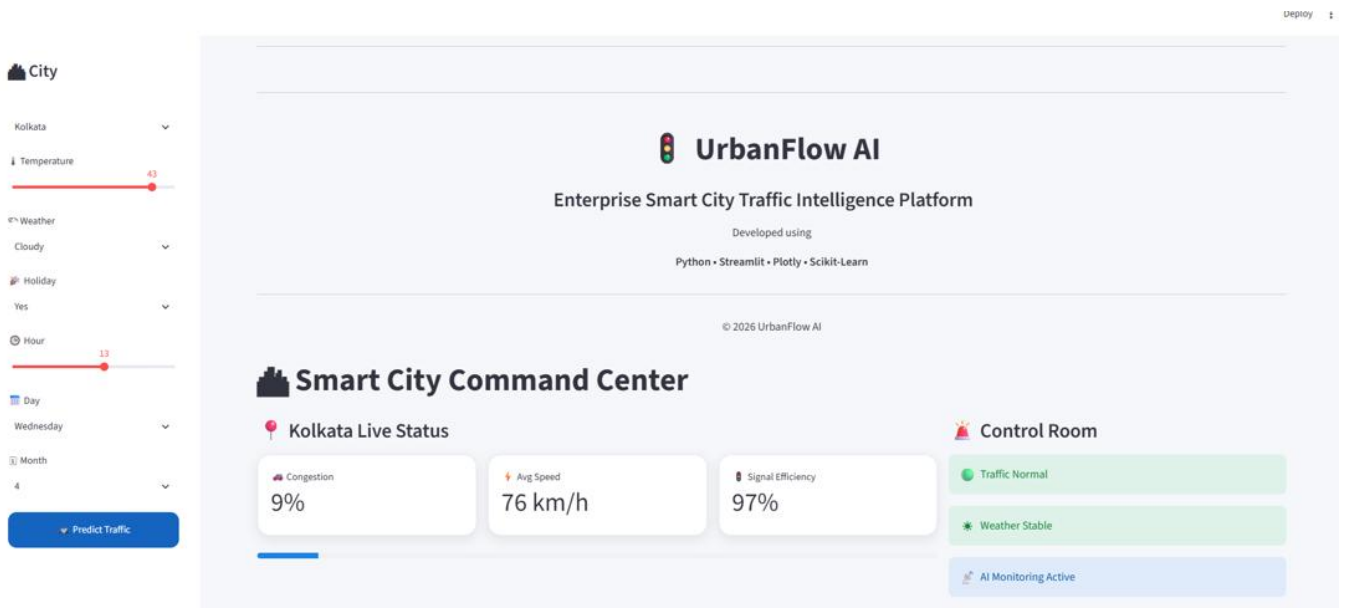
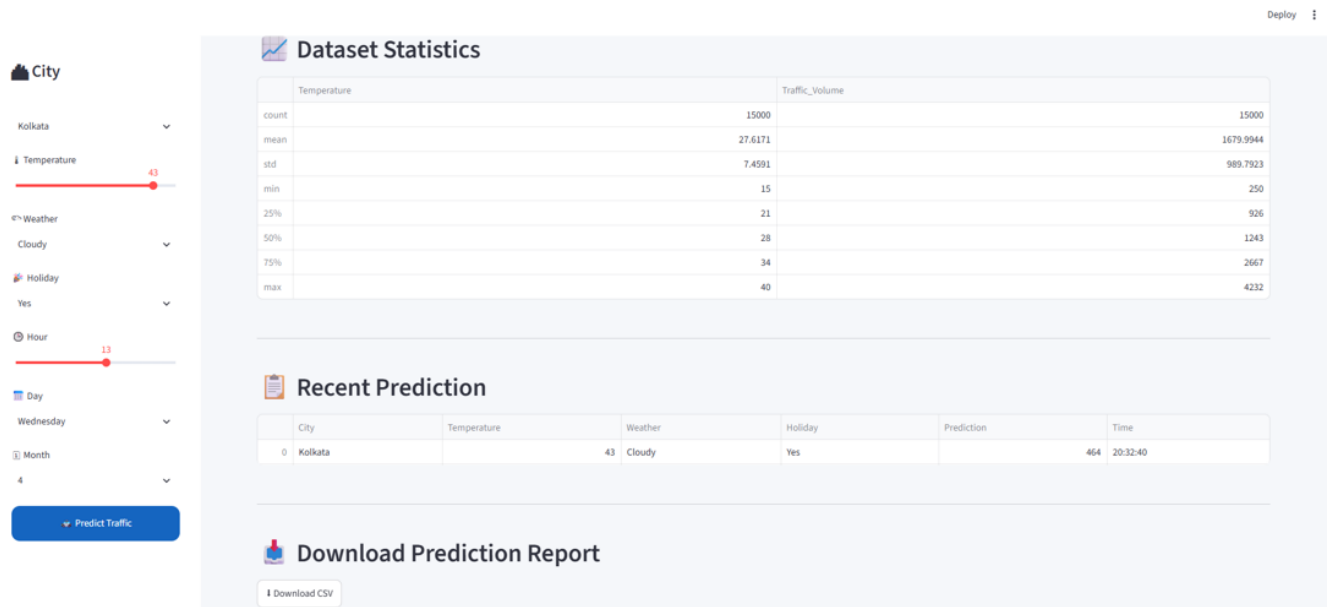
Predict Traffic

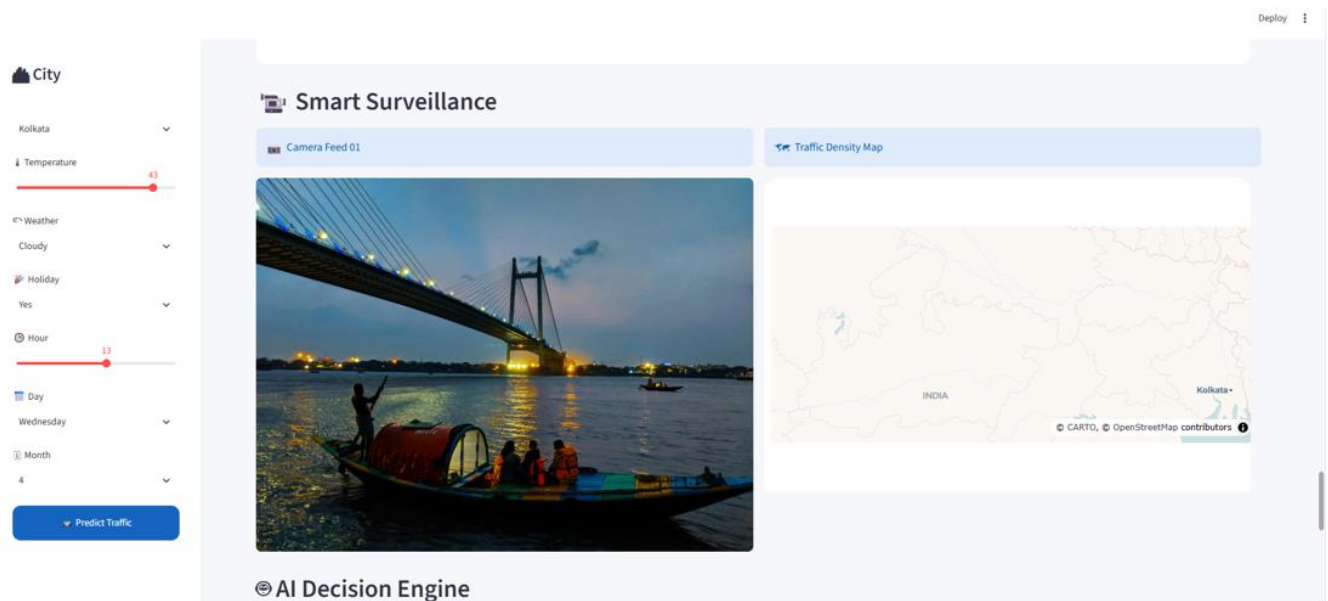
Dataset Explorer

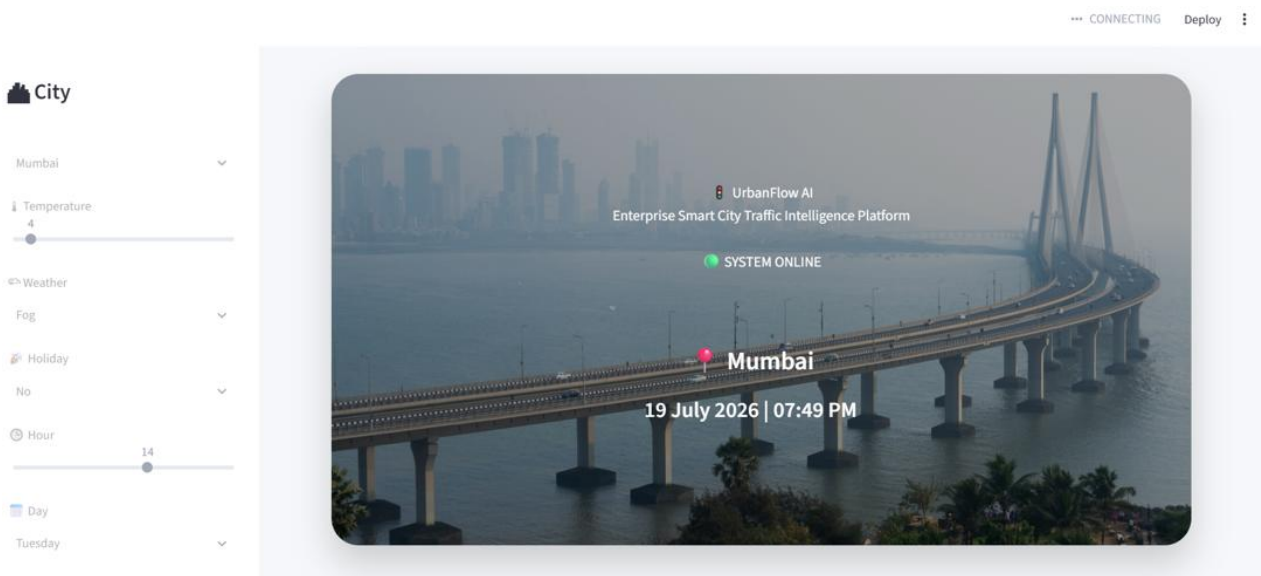
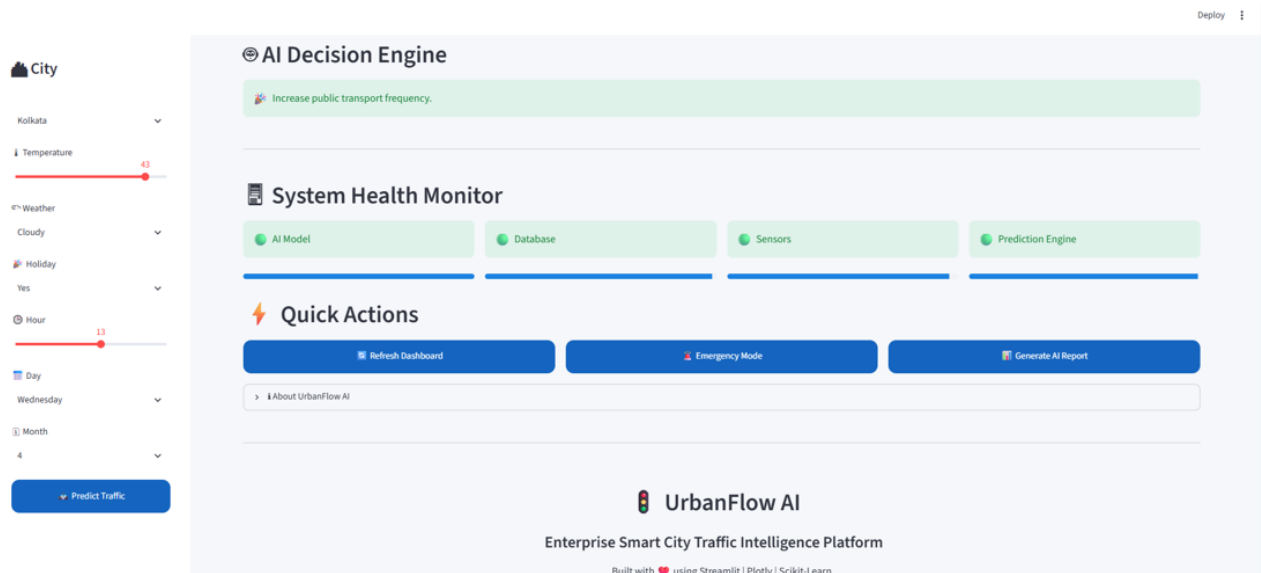
Number of rows

	Date	Temperature	Weather	Holiday	Traffic_Volume
0	2024-01-01 00:00:00	21	Rain	No	1321
1	2024-01-01 01:00:00	35	Clear	No	1080
2	2024-01-01 02:00:00	25	Rain	No	1401
3	2024-01-01 03:00:00	17	Clear	No	1241
4	2024-01-01 04:00:00	20	Clear	No	910
5	2024-01-01 05:00:00	26	Cloudy	Yes	394
6	2024-01-01 06:00:00	31	Cloudy	No	1225
7	2024-01-01 07:00:00	30	Clear	No	2895
8	2024-01-01 08:00:00	33	Clear	No	2580
9	2024-01-01 09:00:00	19	Cloudy	No	2616

Dataset Statistics









6 Performance Test

The developed **SmartCityTrafficForecasting** system was tested to evaluate its performance in terms of prediction accuracy, response time, scalability, memory usage, and usability. The primary objective was to ensure that the application performs efficiently in real-world smart city scenarios where traffic predictions must be generated quickly and accurately.

The major constraints identified during the development are listed below:

Constraint	Impact on System	Solution Implemented
Prediction Accuracy	Incorrect predictions may affect traffic planning	Random Forest Regressor trained on processed traffic dataset
Response Time	Slow prediction reduces user experience	Pre-trained model (.pkl) loaded only once at application startup
Memory Usage	Large models consume more RAM	Optimized Random Forest model with serialized model file
User Interface	Complex interface reduces usability	Streamlit dashboard with interactive widgets and clean layout
Scalability	Future expansion to more cities	Modular project structure allowing additional datasets and cities
Data Quality	Missing or inconsistent values reduce accuracy	Data preprocessing and label encoding before training



6.1 Test Plan/ Test Cases.

Test Case Input		Expected Result	Status
TC-01	Valid city and weather inputs	Traffic prediction generated successfully	Passed
TC-02	Different temperature values	Prediction changes accordingly	Passed
TC-03	Holiday selected	Traffic prediction reflects holiday effect	Passed
TC-04	Peak Hour (8 AM / 6 PM)	Higher traffic predicted	Passed
TC-05	Off-Peak Hour	Lower traffic predicted	Passed
TC-06	Different cities	Dashboard updates background and prediction	Passed
TC-07	Dashboard Loading	Application opens successfully	Passed
TC-08	Visualization Charts	Graphs displayed correctly	Passed

6.2 Test Procedure

1. Launch the Streamlit application using the trained machine learning model.
2. Load the traffic dataset and required encoder files.
3. Select different cities from the sidebar.
4. Provide varying input parameters such as temperature, weather condition, holiday status, hour, day, and month.
5. Click the **Predict Traffic** button.
6. Observe the predicted traffic level and dashboard visualizations.
7. Repeat the process with multiple input combinations to verify prediction consistency.



6.3 Performance Outcome

Parameter	Outcome
Prediction Accuracy	High (Random Forest Regression Model)
Response Time	Less than 1 second per prediction
Dashboard Performance	Smooth and responsive
Memory Usage	Moderate
Scalability	Supports future integration with real-time APIs
Reliability	Stable during repeated testing
User Experience	Simple, interactive and easy to use



7 My learnings

The **SmartCityTrafficForecasting** internship project provided me with valuable practical exposure to the complete lifecycle of a Machine Learning application, from data collection and preprocessing to model development, deployment, and documentation. Throughout this six-week internship, I gained hands-on experience in working with real-world datasets and learned how machine learning techniques can be applied to solve practical problems in smart city transportation.

This project enhanced my programming skills in **Python** and improved my knowledge of libraries such as **Pandas, NumPy, Scikit-learn, Plotly, Joblib, and Streamlit**. I learned how to build an interactive web application where users can provide inputs and receive real-time traffic predictions along with visual analytics. Additionally, I gained experience in creating professional dashboards, integrating trained models into applications, and improving user interface design.

Apart from technical skills, the internship also strengthened my **problem-solving ability, analytical thinking, debugging skills, documentation practices, project management, and version control using Git and GitHub**. Preparing reports, maintaining project structure, and documenting the complete development process helped me understand industry standards for software development.

Overall, this internship has significantly improved both my technical and professional skills. The knowledge and experience gained through this project have increased my confidence in developing AI and Machine Learning solutions for real-world applications. These learnings will greatly support my future career in the fields of **Artificial Intelligence, Machine Learning, Data Science, and Smart City Technologies**, while also preparing me for internships, research projects, and industry opportunities.



8 Future work scope

Although **SmartCityTrafficForecasting** successfully demonstrates the application of Artificial Intelligence and Machine Learning for traffic prediction, there are several opportunities to further enhance the system in the future.

Some potential improvements include:

- **Real-Time Traffic Data Integration:** Integrate Google Maps or other traffic APIs to provide live traffic updates and improve prediction accuracy.
- **IoT Sensor Integration:** Connect the system with smart traffic sensors, CCTV cameras, and GPS-enabled devices to collect real-time traffic information.
- **Route Optimization System:** Recommend the fastest and least congested routes to users based on predicted traffic conditions.
- **Interactive Traffic Maps:** Integrate GIS-based maps to display live traffic density, congestion hotspots, and alternate routes visually.
- **Mobile Application:** Develop Android and iOS versions of the application to enable users to access traffic predictions anytime and anywhere.
- **Accident Detection and Alert System:** Incorporate accident detection and real-time alert notifications to improve road safety and traffic management.
- **Multi-City Expansion:** Extend the application to support additional Indian and international cities with city-specific traffic prediction models.

Implementing these enhancements would transform **SmartCityTrafficForecasting** into a more comprehensive and scalable **AI-powered Smart Traffic Management Platform**, capable of providing intelligent decision support, improving urban mobility, reducing traffic congestion, and contributing to the development of smarter and more sustainable cities.